HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and maintained **Java Spring Boot** REST APIs for internal financial services clients, facilitating data exchange across multiple enterprise systems.
- Refactored inefficient relational database queries and implemented strategic indexing on **PostgreSQL**, reducing analytical report execution times by **34%** across core ERP database clusters.
- Engineered Python scripts to automate ETL processes for scheduled reporting, consolidating data from disparate sources into a central data warehouse for business intelligence teams.
- Designed and implemented **trading and operations-related dashboards** and data tools using Python and SQL to support monitoring, analysis, and decision-making for automated trading systems.
- Contributed to **regulatory reporting** and audit requests by preparing and maintaining documentation for internal controls and compliance procedures.
- Optimized operational processes by identifying manual activities and developing **automation solutions**, improving efficiency and strengthening controls for the trading desk.
- Containerized applications using **Docker** and deployed them to AWS EC2 instances, ensuring consistent environments and facilitating streamlined CI/CD pipelines with Jenkins.

SKILLS

Programming Languages – Python, Java, SQL, R

Frameworks & Libraries – Spring Boot, pandas, NumPy, scikit-learn, Flask, FastAPI

Databases – PostgreSQL, MySQL, Oracle

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git

Tools & Platforms – Power BI, Tableau, Splunk, Dynatrace, Jira, Confluence, Linux

Domain Knowledge – Algorithmic Trading, Financial Markets, Regulatory Reporting, Operational Dashboards, Process Optimization

PROJECTS

Automated Trading System Monitoring Dashboard

- Developed a Python-based application to ingest real-time data from trading systems, providing critical operational metrics for system health.
- Designed and built interactive dashboards using **Power BI** and **SQL** to visualize trade execution status, latency, and system alerts, enabling proactive issue resolution.
- Implemented automated data refresh mechanisms and alerting rules, reducing manual monitoring effort by an estimated **40%** for the operations team.

Financial Transaction Data Pipeline

- Engineered a data pipeline using **Python** and **SQL** to process and transform large volumes of financial transaction data from multiple exchanges.
- Developed data quality checks and validation scripts to ensure the accuracy and integrity of processed data for downstream analysis and reporting.
- Integrated the pipeline with **AWS S3** for scalable data storage and retrieval, supporting ad hoc analysis by quantitative analysts.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)